

Guidance for Paper Review, Presentation Preparation, and Peer-Feedback

Note: Please read this document carefully but understand that it offers a set of guiding principles not requirements.

1. PDF Annotation Guidelines

Your annotated PDF serves as the foundation for your presentation and demonstrates your understanding of the paper.

1.1 What to Include

Your annotations should reflect **active reading**, including:

- **Summaries** of each section or major idea.
- **Clarifications:** restate equations, methods, or concepts in your own words.
- **Questions:** note uncertainties, ambiguous assumptions, or missing details.
- **Connections** to prior knowledge (e.g., relation to policy gradients, Q-learning, credit assignment strategies).
- **Critiques:** identify limitations, questionable assumptions, or experimental gaps.
- **Visual tags:** highlight key figures, equations, or tables and underline important results.

1.2 Tips for Effective Technical Annotation

- Write short, focused comments—pretend you're leaving notes for your future self.
- Paraphrase equations in words (e.g., meaning of each term, intuition).

- Mark connections to DRL paradigms (e.g., actor–critic, off-policy learning, exploration mechanisms).
- Annotate figures describing what they *mean*, not just what they *show*.

2. Presentation Guidelines

Your presentation should teach the paper to the class with clarity and depth.

2.1 Goals of the Presentation

- Accurately convey the **problem, method, theoretical contributions**, and **results**.
- Explain *why* the contribution matters in DL/RL research and application.
- Help your peers develop practical and intuitive understanding.
- Critically evaluate the work.

2.2 Recommended Presentation Structure (feel free to adapt for maximum effect)

1. Title slide

- Paper title, year, authors, venue, your group names

2. Motivation

- What problem does this paper solve?
- Why is it important in DL/RL research / application?

3. Background / Preliminaries

- DL/RL fundamentals required to understand the method
- Key definitions, notations, or equations

4. Contribution Summary

- What is new or significant?
- (Try to summarize in 2–3 bullet points)

5. Method / Approach

- Clear explanation of the architecture, algorithm, and/or theoretical framework
- Step-by-step walk-through of the algorithm
- Visual aids are especially helpful

6. Experiments

- Experimental setup
- Environment(s), baseline comparisons, metrics

7. Key Results & Interpretation

- Explain results in a conceptual way
- Interpretation > description
- Discuss ablations, if available

8. Strengths & Limitations

- Strong points of methodology and evaluation
- Weaknesses, potential biases, generalization concerns

9. Discussion / Open Questions

- What research directions does this paper inspire?
- How might others extend or challenge it?

10. Conclusion

- What should the audience take away?

2.3 Style Expectations

- 12–15 minutes presentation + time for questions
- Clean slides (avoid text-heavy content)
- Prefer diagrams > text
- Use consistent fonts, colours, labels
- When presenting equations, explain intuition first
- Credit all figures taken from the paper (and elsewhere)

3. Peer Review Guidelines

Each group is paired with another review group. You are expected to give **constructive, actionable** feedback.

3.1 Components of Peer Review

You will review the partner group's:

1. **Annotated PDF**
2. **Presentation Draft**

3.2 How to Give Constructive Feedback

Your feedback should be:

- **Specific:** reference a precise slide or annotation
- **Actionable:** say what could be improved and how
- **Respectful:** assume good intentions
- **Balanced:** highlight strengths and not only weaknesses

3.3 Suggested Feedback Framework (feel free to adapt for maximum effect)

A. Summarize the work as you understand it

- Provide a short summary of the paper after reviewing the annotations and slides (the supporting group is not required to read the paper.) This will help the presenters identify any gaps in their presentation of the material.

B. Content Accuracy & Depth

- Are definitions and equations correct?
- Does the presentation accurately reflect the paper's method?
- Are connections to DL/RL background clearly explained?
- Are assumptions or limitations correctly identified?

B. Clarity of Explanation

- Do slides effectively communicate key ideas?
- Are complex concepts explained intuitively?
- Is the narrative coherent and well structured?
- Are the motivations and contributions clear?

C. Technical Understanding

- Does the group demonstrate mastery of the material?
- Do annotations reflect thoughtful engagement with the paper?
- Are questions or unclear points identified appropriately?

D. Visual Presentation Quality

- Visual flow: Are slides organized logically?
- Readability: font size, spacing, colour contrast.
- Figures/diagrams: annotated, explained clearly.

E. Suggestions for Improvement

Examples:

- “Slide 5: Consider adding intuition behind this equation.”
- “Try reducing text on slide 3; the main idea can be summarized in 2 bullets.”

- “Add a diagram showing the model architecture.”
- “Clarify why baseline X is relevant in the experiments.”

5. Helpful Resources

Below are reliable resources to support your reading, annotation, and presentation preparation.

Research Paper Reading & Reviewing

- *How to Read a Paper* (Keshav):
<https://web.stanford.edu/class/cs245/readings/how-to-read-a-paper.pdf>.
- NEURIPS Review Guidelines: <https://neurips.cc/Conferences/2025/ReviewerGuidelines>
- ICML Reviewer Tutorial: https://drive.google.com/file/d/15hPTA64h31ShaoybLWeU3moZan7zVbr_/view
- AISTATs Reviewer guidelines: https://aistats.org/aistats2024/reviewer_guidelines.html

Presentations & Scientific Communication

- *How to Give a Good Research Talk* (Simon Peyton Jones):
<https://www.microsoft.com/en-us/research/academic-program/give-great-research-talk/>
- In the class presentation, you may go into more detail than is suggested here. Part of your task is to teach us about the paper. Nonetheless, this very useful to read for your class presentation and future conference presentations.
- MIT Technical Communication resources:
<https://tll.mit.edu/teaching-resources/how-to-teach/presentation-tips/>